

Calculators, cell phones, or notes are not allowed.

- (1) Given that 2 is a primitive element of $(\mathbb{Z}/37\mathbb{Z})^*$ find a primitive 3-rd root of 1 in $\mathbb{Z}/37$ and a square root of -3 (hint: in $\mathbb{C}$ let $\zeta = \exp(2\pi i/3) = -1/2 + \sqrt{-3}/2$ be a primitive 3-rd root of 1, then $\zeta - \zeta^2 = \sqrt{-3}$).

$2^5 = 32 = -5$, $2^{10} = 25$, $2^{12} = 100 = 26$ is a primitive 3-rd root of 1 in $\mathbb{Z}/37\mathbb{Z}$ (since $12 = \frac{1}{3} |(\mathbb{Z}/37\mathbb{Z})^*|$).

Let $\zeta = 26 = -11$ then $\zeta^2 = 121 = 10$ so $\zeta - \zeta^2 = 16$.

Alternatively you may have noticed that $2^4 = 16$, $2^8 = 256 \cong_{37} -3$.

- (2) Does $X^2 - 6X + 15 = 0$ have a solution modulo 61?

Completing the square we get $(X - 3)^2 = -6$ and so we must check if -6 is a square modulo 61. We have $(\frac{-6}{61}) = (\frac{-1}{61})(\frac{2}{61})(\frac{3}{61})$. Now $(\frac{-1}{61}) = 1$ as $61 \cong_4 1$, $(\frac{2}{61}) = -1$ as $61 \cong_8 5$. Since $61 \cong_3 1$, then $(\frac{3}{61}) = (\frac{61}{3}) = (\frac{1}{3}) = 1$ hence $(\frac{-6}{61}) = -1$ so -6 is not a square modulo 61 and the equation can not be solved.

Alternatively you could have computed $(-6)^{30} = 6^{30}$. We have $6^2 = 36 = -25$, $6^4 = 625 = 15$, $6^8 = 225 = 42 = -19$, $6^{16} = -5$, $6^{30} = -5 \cdot -19 \cdot 15 \cdot -25 = 95 \cdot -375 = 34 \cdot -9 = -306 = -1$.